Teamwork Agreement Template

For the MathWorks Classroom Challenge Projects

Project: Image-Based Defect Detection for Manufacturing Inspection (IDBBMI)

Team Name: Team 7

Instructor: Kevin Graham

Program/Class: Engineering Pathways Program

Members and Roles

(*Roles are contingent and may be changed at any point throughout the duration of this project***)**

1. Ryan Rau - Quality Assurance & Reproducibility Lead
2. Isaac Alvarado - Analysis/Validation Lead
3. Dayana Pascual Sanchez - Project Manager/Modeling Lead
4. Kaitlyn Lopez - Documentation & Visualization Lead

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: Discord (Team7 Server)

- Secondary channel: Email

- Response time: 24 hours on weekdays and 48 hours on weekends

- Meeting cadence: Wednesdays 2-3pm, 2-hour meeting over the weekend (time to be determined each week), additional meeting times may be created with written consent from all team members.

- Decision records: Project Manager logs major decisions in the Meeting Notes document. Decided upon tasks are assigned/updated in the Task List by Project Manager before the end of each meeting.

4) Collaboration & Tools

- Version control: https://github.com/rbradleyrau-creator/Image-Based-Defect-Detection-for-Manufacturing-Inspection-Team7

- Data management: All datasets, parameters, and assumptions documented in README.md. All matlab scripts inside of the matlab script folder on Github.

File naming & style: Pascal Case, snake case for version numbers.

- Backups: Project Manager keeps old versions numbered inside of the github as backups before major changes.

5) Quality Standards (Definition of 'Done')

An item is considered to be done when it has been documented (references, assumptions, units, specified in comment code, etc) and meets the requirements described in either the project description or task list. For Matlab scripts, the code must run from README.md, be tested with edge cases, and have been verified by another person. For final project items, All members must be well informed and knowledgeable with the item in question for it to be considered finalized and complete.

6) Timeline & Milestones

- Week 1: (July 5th - July 11th)
  - Create teamwork agreement
  - Go through recommended training as deemed necessary.
  - Problem Framing, determining dataset(s) and parameters
  - Read relevant documentation to key MATLAB functionality (i.e. imageDataStore, resnet18, etc.)
  - Organize dataset (choose a dataset, define defect classes and labels, document in the github)
  - Have a single-frame evaluation model done.

- Week 2: (July 12th - July 18th)

- Integrate an AI Classifier to the single-frame model (classifies PASS/FAIL with a confidence rating)
- Integrate both classic evidence and AI decisions into a single system.
- Perform batch testing using the new integrated model and generate initial confusion matrix, report yield, and defect rates.
- Results from a robustness test where images were altered using changes to brightness/contrast, blur, and noise.

- Week 3: (July 19th - July 25th)
  - Optimize and Improve inspection model to meet project requirements.
  - Validation (ensure code functions, all cases tested, output well documented/comments)
  - Quality assurance (clean up Matlab code, ensure functionality in README.md, etc.)
  - Add live script to model(s) to clearly illustrate classification(s).
  - Report outline written and section tasks assigned before the start of Week 4.
  - If time, write final report(s) on current model(s)

- Week 4: (July 26th - August 1st)
  - Write/finalize report(s) on current model(s)
  - Make adjustments to MATLAB code (i.e. comments, code structure, spacing, etc.)
  - Study working model(s) in preparation for final project interview/proposal.
  - Create/practice presentation on working model(s)
  - Cleanup Repository and ensure everything has been properly documented
  - AI-Based Segmentation (if time, must be agreed upon by entire group)

7) Decision-Making & Conflict Resolution

- Default: Consensus after Discussion

- If blocked >48 hrs: Majority vote, Project Manager breaks ties.

- Technical disagreements: Evidence-based comparison before vote

- Escalation: If unresolved, consult instructor/mentor with a concise memo during office hours.

8) Workload, Accountability & Attendance

- Task board: To-Do / Doing (name) / Review (reviewers name) / Done (fully documented and checked off in task list)

- Accountability: Missed deadline → recovery plan within 24 hrs; repeated missed deadlines, trigger role redistribution.

- Attendance: If you'll miss a meeting, give 24-hour notice alongside the reason, provide an alternative time (if able), and share updates asynchronously. Keep members up to date with information and tasks from meetings. Repeatedly missing meetings, may trigger role/task redistribution (at discretion of PM).

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- Well-structured Github repository with version history, clear README, and well documented code.

 - A single-image inspection function that returns a PASS/FAIL AI classification decision, a confidence score in the PASS/FAIL decision, and the evidence overlay and metrics to support the AI decision

 - A script for a batch test runner that that divides all data into a training set and a testing set or uses cross-validation, evaluates the trained AI model, and produces a confusion matrix and yield/defect summary

 - Results from a robustness test where images were altered using changes to brightness/contrast, blur, and noise

- Final Report summarizing various sections such as our methods and results

- Presentation using findings from report

12) Agreement & Signatures

Name & Signature: Isaac Alvarado

Date: 7/5/2026

Name & Signature: Kaitlyn Lopez

Date: 7/5/2026

Name & Signature: Dayana Pascual Sanchez

Date: 7/5/2026

Name & Signature: Ryan Rau

Date: 7/5/2026